

The Synergy Reality Test — Practitioner’s Guide

How to run the analysis: the rent split, the entry floor, worked end-to-end

Methods companion to the one-pager — for analysts (**internal**)

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What this is. The one-pager *sells* the Synergy Reality Test; this guide *runs* it. It is what you reach for when a client (or a skeptical colleague) asks “show me how you split durable from fragile and where the floor comes from.” Same roll-up as the one-pager, carried through end to end.

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1 The principle: a roll-up captures rent, it does not create it

The governing fact is conservation. The total economic rent in a market is fixed; consolidation only *moves* it between pools. For J symmetric competitors the unit of market rent splits three ways:

$$\underbrace{\frac{1}{J}}_{\text{protected baseline}} + \underbrace{\rho \frac{J-1}{J}}_{D: \text{dissipated by competition}} + \underbrace{(1-\rho) \frac{J-1}{J}}_{K: \text{curvature shield (variety)}} = 1.$$

D is rent competition wastes (redundant overhead, price wars); K is surplus protected by product variety/differentiation; $1/J$ is the monopoly baseline. The dial $\rho \in (0,1)$ is the *substitutes-vs-complements* regime: high ρ = mostly wasteful competition, low ρ = mostly genuine variety. A roll-up runs two checks off this identity.

2 Step 0 — Define the market

Fix the relevant antitrust/economic market and count the effective competitors J . Synergy is measured *within this market*; widen or narrow the boundary and the numbers move. *State it.*

3 Check 1 — Durable vs. fragile (the ρ split)

3.1 The mechanics

Consolidating one competitor, $J \rightarrow J - 1$, moves rent into the protected baseline:

$$\Delta\left(\frac{1}{J}\right) = \frac{1}{J-1} - \frac{1}{J} = \frac{1}{J(J-1)} \quad (\text{captured}),$$

sourced, because the total is conserved, as

$$\underbrace{\frac{\rho}{J(J-1)}}_{\text{recaptured waste } (-\Delta D) = \text{durable}} + \underbrace{\frac{1-\rho}{J(J-1)}}_{\text{destroyed variety } (-\Delta K) = \text{fragile}}.$$

So of *every* dollar of captured synergy, fraction ρ is **durable** (recaptured waste — a real efficiency gain that takes nothing from anyone) and fraction $(1-\rho)$ is **fragile** (variety converted to rent — a transfer from customers, the antitrust-relevant part, which invites entry and reverts). The split is J -independent; only its *size* $1/(J(J-1))$ depends on J .

3.2 Estimating ρ — the one judgment that drives everything

ρ is exogenous and it *is* the analysis. Estimate it as the share of the projected synergy that is genuine cost/efficiency recapture (consolidating duplicated functions, ending value-destroying price competition) versus pure pricing power from fewer rivals. Anchor it: scrutinize the synergy model line by line, classify each item as efficiency or price-from-concentration, and let ρ be the efficiency share. Document the call.

3.3 Worked example

A platform in a market of $J = 6$ pitches consolidation to 3, projecting **\$60M** of EBITDA synergy — which, being margin from consolidation, is captured rent in full. We estimate the durable share at $\rho = 0.60$ (60% efficiency recapture, 40% competition reduction):

Projected synergy	\$M	Character
Efficiency recaptured ($\rho = 0.60$)	36	durable — sticks
Competition-reduction rent ($1 - \rho = 0.40$)	24	fragile — reverts
Captured synergy	60	

\$36M durable; \$24M fragile. Price the platform on the \$36M.

4 Check 2 — The entry floor (how far you can hold it)

4.1 The mechanics (Salop)

A differentiated market with circular product space of differentiation t and entry cost f has equilibrium per-firm markup and free-entry count

$$\mu(n) = \frac{t}{n}, \quad n^* = \sqrt{t/f}, \quad \mu(n^*) = \sqrt{tf} \text{ (the entry-proof markup),} \quad \text{durable HHI} = \frac{1}{n^*} = \sqrt{f/t}.$$

Markup rises as players fall ($\mu = t/n$), and a merger lifts it by $\Delta\mu = t/(J(J-1))$. But n^* is the level entry restores: consolidating *below* n^* pushes markup above $\sqrt{tf}$, and that excess is **fragile** — it draws entry back to n^* .

4.2 Worked example

With differentiation $t = \$20\text{M}$ and entry cost $f = \$1.25\text{M}$:

$$n^* = \sqrt{20/1.25} = \sqrt{16} = 4 \text{ players,} \quad \mu(n^*) = \sqrt{20 \cdot 1.25} = \$5\text{M,} \quad \text{durable HHI} = 1/4 = 0.25.$$

Consolidation step	Character
6 $\rightarrow$ 4 players	durable — down to the natural floor $n^* = 4$
4 $\rightarrow$ 3 players	fragile — below n^* ; entry restores ~ 4 and competes the gain away

The two checks are complementary, never added. Check 1 sizes the fragile rent in dollars (\$24M, via ρ); Check 2 locates it in the deal sequence (the $4 \rightarrow 3$ step, via the floor). They are different lenses on the same fragile rent — one in \$, one in deal count — and are *not* summed. (Converting Check 2's per-unit markup into total dollars would need a volume input, which is why they stay separate.)

5 “How far should we consolidate?” — and why there is no forced optimum

Netting the recaptured efficiency against the destroyed variety gives the per-merger value (in the curvature metric)

$$\frac{\rho}{J(J-1)} - \frac{1-\rho}{J(J-1)} = \frac{2\rho-1}{J(J-1)}, \quad \text{cumulative } J \rightarrow n : (2\rho-1) \Delta\text{HHI}.$$

The sign is fixed by $(2\rho-1)$ and the magnitude *grows* as $J \rightarrow 2$ — increasing returns, not diminishing. So the geometry alone gives a **corner solution**: consolidate fully if $\rho > \frac{1}{2}$, not at all if $\rho < \frac{1}{2}$. *There is no interior “optimal number of acquisitions” in the math.* An interior stopping point appears only when you **feed** a rising cost — antitrust hazard or integration dis-synergy — or impose the entry floor of Check 2. The defensible recommendation is therefore the *durable ceiling* (n^*), not a profit-maximizing k : in the example, $\rho = 0.6 > \frac{1}{2}$ says “keep going” on pure rent, but the floor says everything below 4 players is fragile, so 4 is the durable stopping point and the $4 \rightarrow 3$ deal buys rent that reverts.

6 Judgment checkpoints (forced vs. fed)

Forced by the algebra	Fed by you (state as assumptions)
The rent identity $1/J + D + K = 1$; merger curvature loss $(1 - \rho)/(J(J - 1))$; HHI = $1/J +$ dispersion.	ρ , the substitutes-vs-complements regime (drives the entire split).
Salop markup t/n ; free-entry $n^* = \sqrt{t/f}$, markup $\sqrt{tf}$.	Differentiation t and entry cost f (locate the floor).
Capture-not-creation (the conserved total).	A price w to put \$ on destroyed variety; the bridge that one market is both ρ - and t -shaped.
Net per-merger value $(2\rho - 1)/(J(J - 1))$.	Antitrust thresholds, integration cost (to get an interior k); the reversion <i>timeline</i> .

Out of scope entirely: the *speed* of reversion. The test gives the durable *level* (a comparative-static fixed point), never how fast entry competes the fragile rent away — flag it as a judgment, never invent a number.

7 Why it works (self-contained)

Each step has an elementary reason — a normalized partition, one line of algebra, and the textbook free-entry condition. Nothing reaches beyond that.

- **The rent decomposition is bookkeeping on a normalized total.** Set total market rent to 1. With J equal competitors the symmetric floor is $1/J$; the rest, $(J - 1)/J$, is contested. Let ρ be the fraction of that contested rent which is wasteful competition (recoverable) and $(1 - \rho)$ the fraction protected by differentiation. The three pieces sum to 1 by construction — it’s a partition, not an assumption. (ρ is the one input you estimate.)
- **The merger move is one line.** $\Delta(1/J) = \frac{1}{J-1} - \frac{1}{J} = \frac{1}{J(J-1)}$ shifts into the protected baseline, of which ρ is durable (recovered waste) and $(1 - \rho)$ is fragile (lost variety). Net per merger = $\frac{2\rho-1}{J(J-1)}$ — positive iff $\rho > \frac{1}{2}$.
- **The entry floor is the free-entry condition.** In the standard circular-market model each firm’s markup is t/n and it serves $1/n$ of the market, so per-firm operating profit is $\frac{t}{n} \cdot \frac{1}{n} = \frac{t}{n^2}$. Free entry drives that to the entry cost f : $\frac{t}{n^2} = f \Rightarrow n^* = \sqrt{t/f}$, with entry-proof markup $t/n^* = \sqrt{tf}$ and durable concentration $1/n^* = \sqrt{f/t}$. Four lines, standard IO.

What’s exact is the split and the floor formula; the inputs (ρ, t, f) are judgment, and there is no “optimal k ” — the geometry gives a corner, so an interior stop needs a fed cost (antitrust / integration). Present it as “exact and transparent,” never as an external proof.

Companion to “The Synergy Reality Test” and to the Value-Creation Bridge guide. Draft prepared by the ENT 105 team for Creo Advisors. Figures are illustrative and reconcile with the one-pager. The rent split and the entry floor are grounded in conservation-based accounting; ρ, t, f , antitrust thresholds, and the reversion timeline are client-specific inputs set at scoping.